67539:
LLMs and AI Agents for Research

Tom Hope

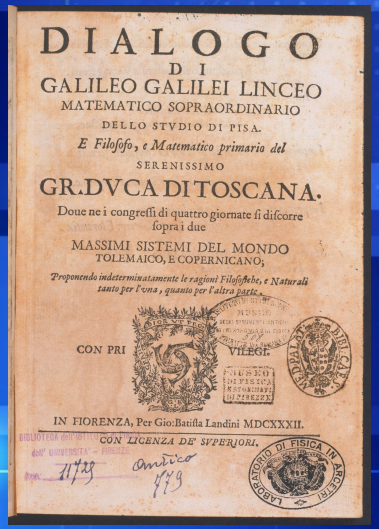
Senior Lecturer, The Hebrew University of Jerusalem
Research Scientist, Allen Institute for AI (AI2/Semantic Scholar)



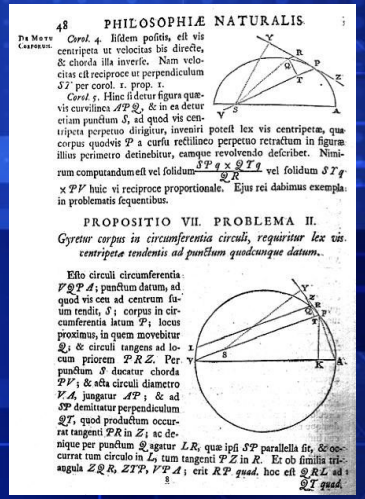


AI2

Until mid-17th century: Self-published books and letters!



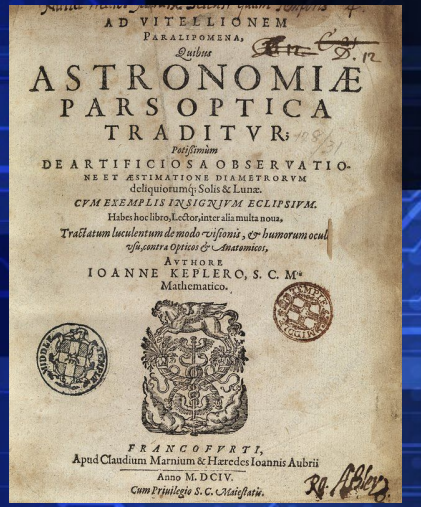
DIALOGO
DI
GALILEO GALILEI LINCEO
MATEMATICO SOPRAORDINARIO
DELLO STVDIO DI PISA.
E Filosofo, e Matematico primario del
SERENISSIMO
GR. DVCA DI TOSCANA.
Doue ne i congressi di quattro giornate si discorre sopra i due
MASSIMI SISTEMI DEL MONDO
TOLEMAICO, E COPERNICANO;
Proponendo indeterminatamente le ragioni Filosofiche, e Naturali tanto per l'vna, quanto per l'altra parte.
CON PRI VILEGI.
IN FIORENZA, Per Gio: Batista Landini MDCXXXII.
CON LICENZA DE' SVPERIORI.



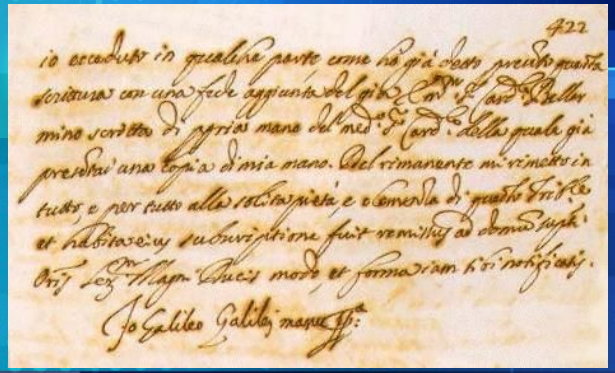
48 PHILOSOPHIÆ NATURALIS
De Motu Corporum.
Corol. 4. Iisdem positis, est vis centripeta ut velocitas bis directæ, & chorda illa inversè. Nam velocitas est reciproce ut perpendiculum *ST* per corol. 1. prop. 1.
Corol. 5. Hinc si detur figura quævis curvilinea *APQ*, & in ea detur etiam punctum *S*, ad quod vis centripeta perpetuo dirigitur, inveniri potest lex vis centripetæ, qua corpus quodvis *P* a cursu rectilineo perpetuo retractum in figuræ illius perimetro detinebitur, eamque revolvendo describet. Nimirum computandum est vel solidum $\frac{SP^4 \times QT^4}{QR}$ vel solidum $ST^4 \times PV$ huic vi reciproce proportionale. Ejus rei dabimus exempla in problematis sequentibus.

PROPOSITIO VII. PROBLEMA II.
Gyretur corpus in circumferentia circuli, requiritur lex vis centripetæ tendentis ad punctum quodcunque datum.

Esto circuli circumferentia *PQP A*; punctum datum, ad quod vis ceu ad centrum fa-tum tendit, *S*; corpus in circumferentia latum *P*; locus proximus, in quem movebitur *Q*; & circuli tangens ad locum priorem *PRZ*. Per punctum *S* ducatur chorda *PV*; & acta circuli diametro *VA*, jungatur *AP*; & ad *SP* demittatur perpendiculum *QT*, quod productum occurrat tangenti *PR* in *Z*; ac denique per punctum *Q* agatur *LR*, quæ ipsi *SP* parallela sit, & occurrat tum circulo in *L*, tum tangenti *PZ* in *R*. Et ob similia triangula *QQR*, *ZTP*, *VPA*; erit *RP quad.* hoc est $\frac{QRL}{QT}$ ad *QT quad.*



AD VITELLIONEM
PARALIPOMENA,
Quibus
ASTRONOMIÆ
PARS OPTICA
TRADITVR:
Potissimùm
DE ARTIFICIOSA OBSERVATIONE ET ÆSTIMATIONE DIAMETRORVM deliquiorumq; Solis & Lunæ.
CVM EXEMPLIS INSIGNIVM ECLIPSIVM.
Habes hoc libro, Lector, inter alia multa noua,
Tractatum luculentum de modo visionis, & humorum oculi, visu, contra Opticos & Anatomicos,
AVTHORE
IOANNE KEPLERO, S. C. Mᵃᵉ Mathematico.
FRANCOFVRTI,
Apud Claudium Marnium & Hæredes Ioannis Aubrii
Anno M. DCIV.
Cum Priuilegio S. C. Maiestatis.



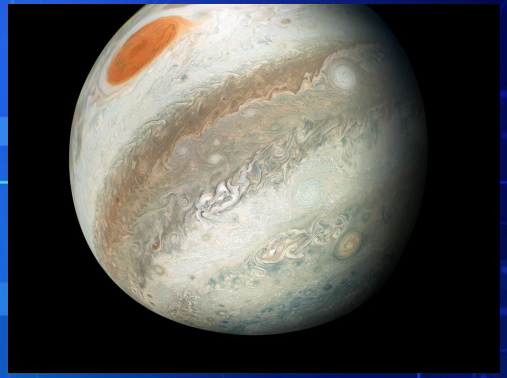
422
io escadure in qualche parte come ho già detto prendo questa
scrittura con una fede aggiunta del già Em.mo Sig.r Card.le Bellar-
mino scritta di propria mano del med.mo Sig.r Card.le della quale già
presentai una copia di mia mano. Del rimanente mi rimetto in
tutto, e per tutto alla retta pietà, e clemenza di questo Tribu-
nale, et habitavi in subiectione fuit promptus ad obedi-
Christ. ex Magn. Ducis mode et forma iam h.ti notissimi.
Io Galileo Galilei mano pp.

PHILOSOPHICAL
TRANSACTIONS:
GIVING SOME
ACCOMPT
OF THE PRESENT
Undertakings, Studies, and Labours
OF THE
INGENIOUS
IN MANY
CONSIDERABLE PARTS
OF THE
WORLD.

Vol I.
For *Anno* 1665, and 1666.

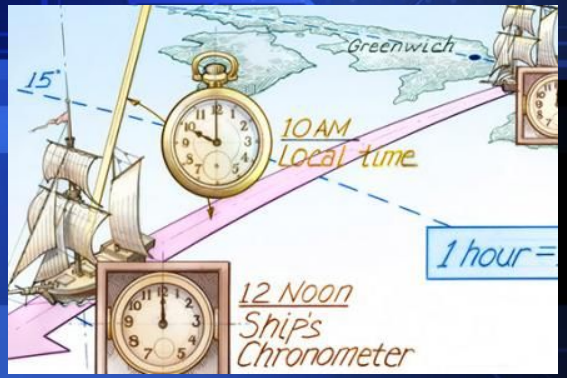
In the *SAVOY*,
Printed by *T. N.* for *John Martyn* at the Bell, a little without *Temple-Bar*, and *James Allestry* in *Duck-Lane*,
Printers to the *Royal Society*.
Presented by the Author May. 30th. 1667.

...1665
first scientific journal!







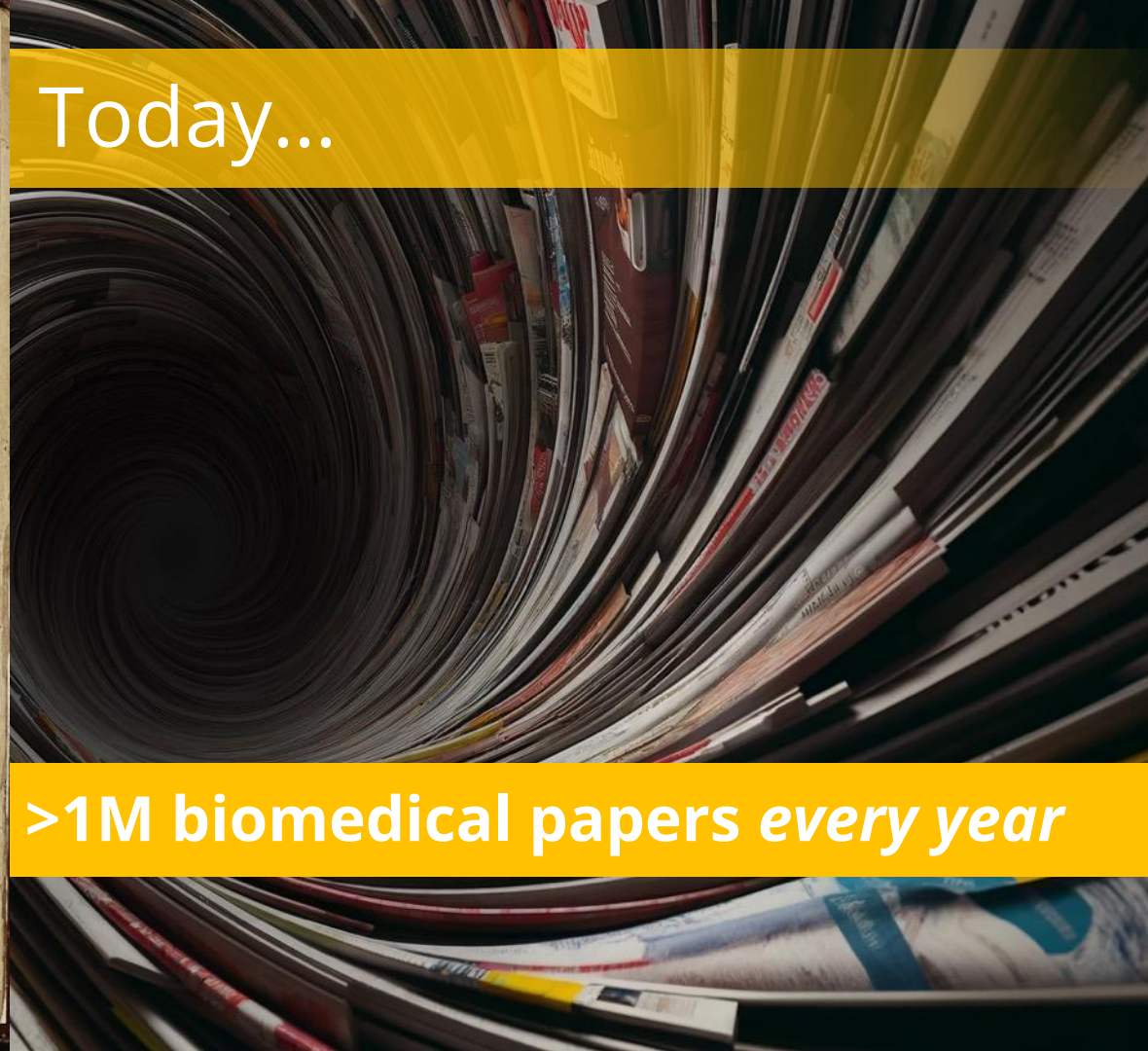


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Presented by the Author May. 30th. 1667.

Today...



>1M biomedical papers *every year*

Explosion of Science



Literature

Data

Tools, resources

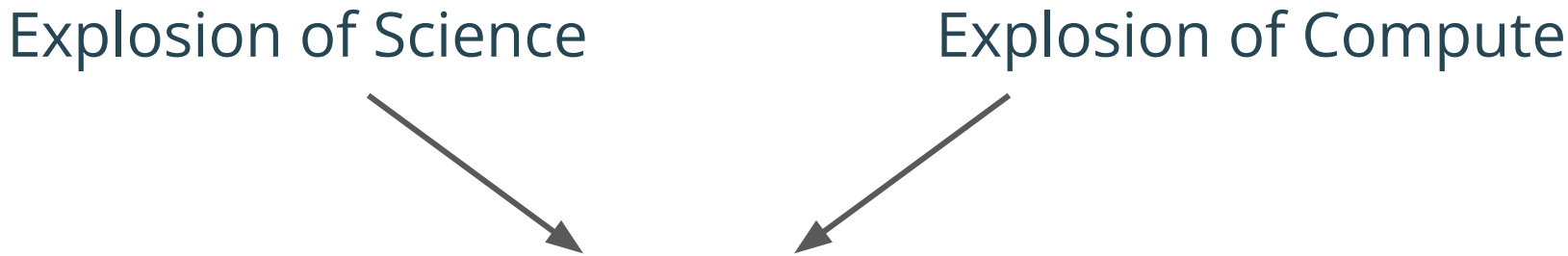
Discussions

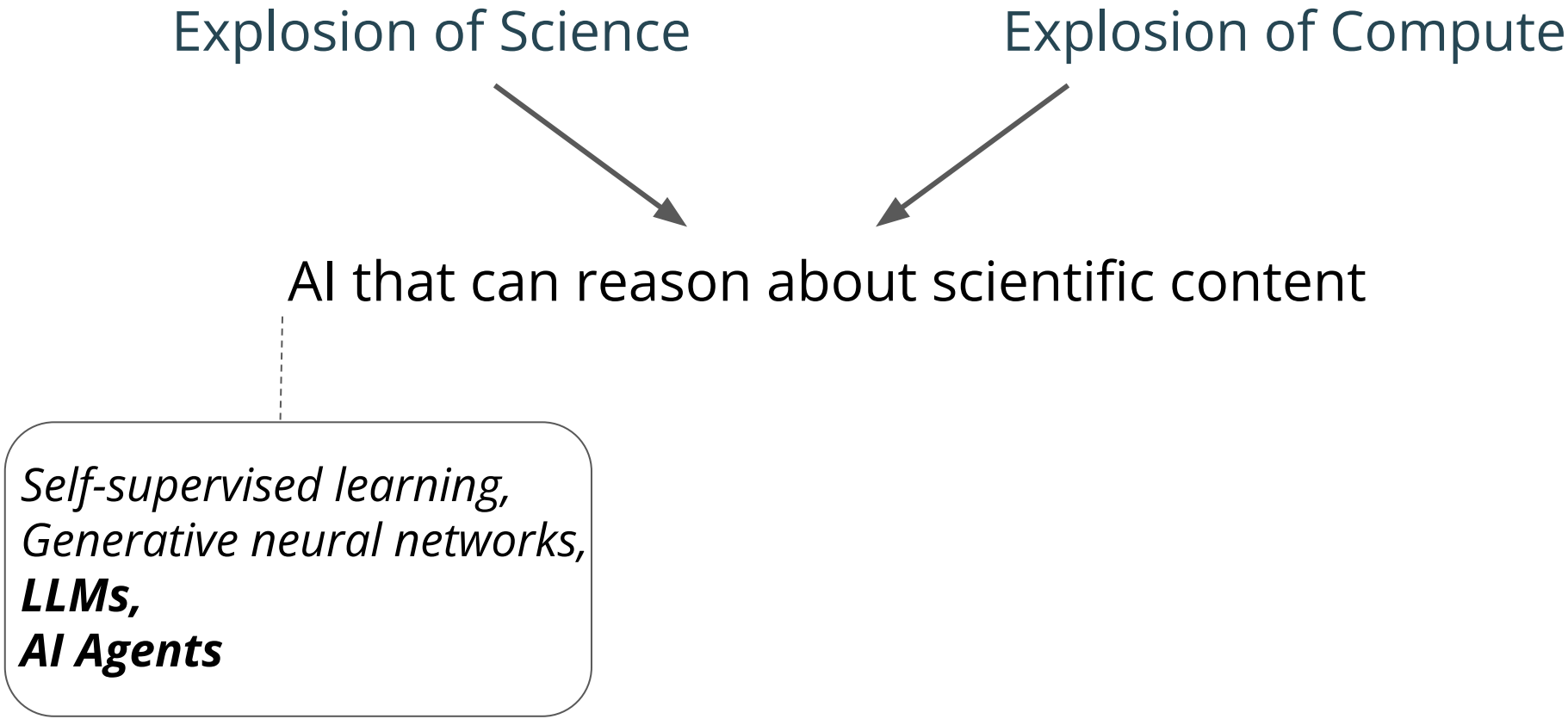
...

All in the digital sphere

Explosion of Compute




Explosion of Science
Explosion of Compute
AI that can reason about scientific content

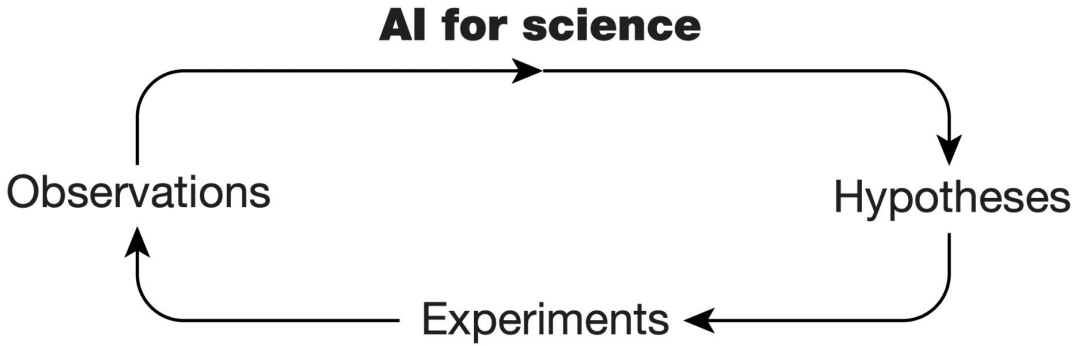

Explosion of Science
Explosion of Compute
AI that can reason about scientific content
Self-supervised learning,
Generative neural networks,
LLMs,
AI Agents

Explosion of Science

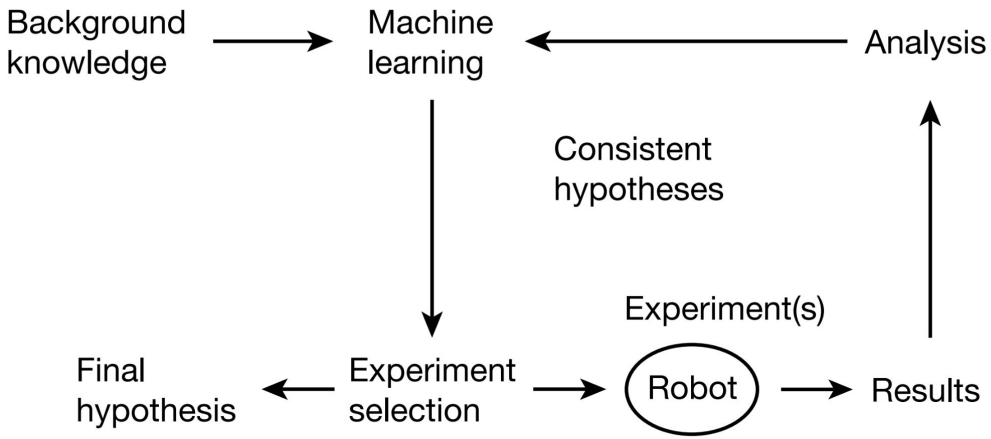
Explosion of Compute

AI that can reason about scientific content

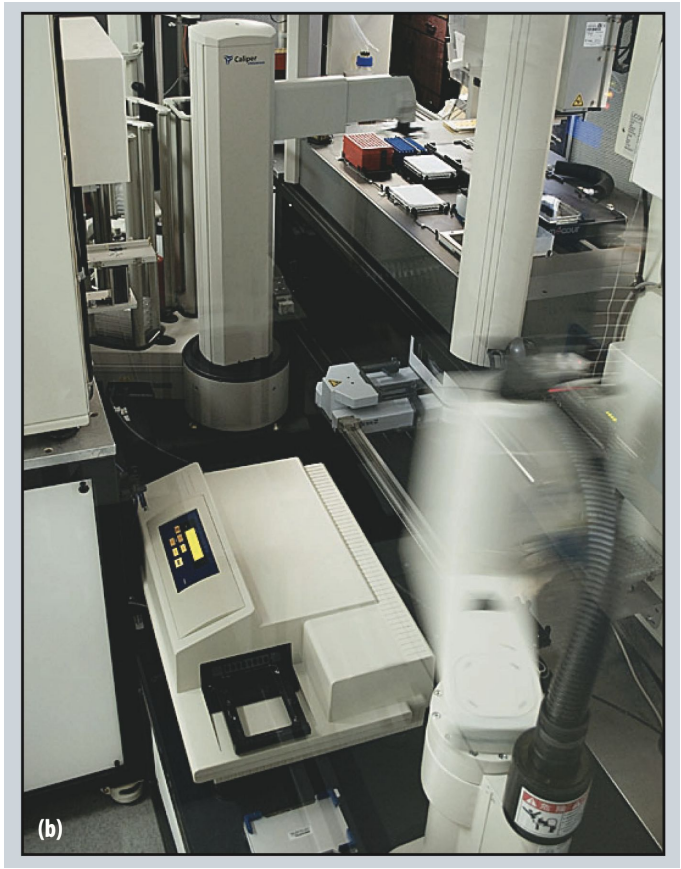
Self-supervised learning,
Generative neural networks,
LLMs,
AI Agents



Autonomous Robot Scientists (2004)



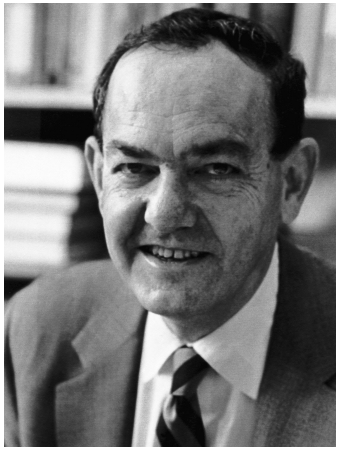
1 The Robot Scientist hypothesis-generation and experimentation loop.



* Ross King, Functional genomic hypothesis generation and experimentation by a robot scientist, Nature, 2004

BAC0N.5: THE DISCOVERY OF CONSERVATION LAWS[1]

Pat Langley
Gary L. Bradshaw
Herbert A. Simon
Department of Psychology
Carnegie-Mellon University
Pittsburgh, Pennsylvania 15213 USA



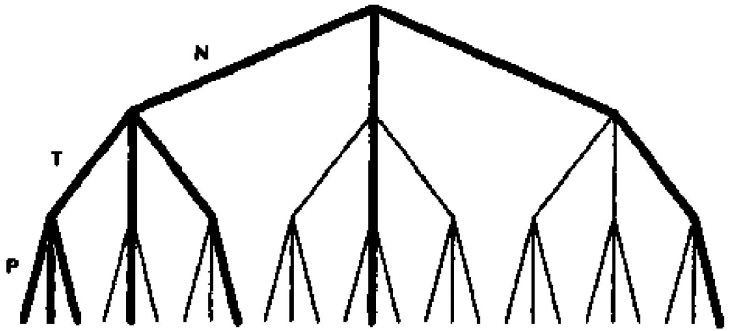


Figure 1. Search trees for the ideal gas law.

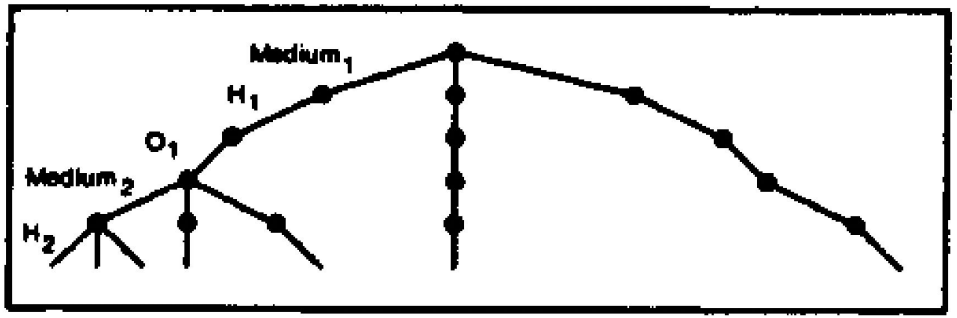


Figure 2. Search tree for Snell's law.

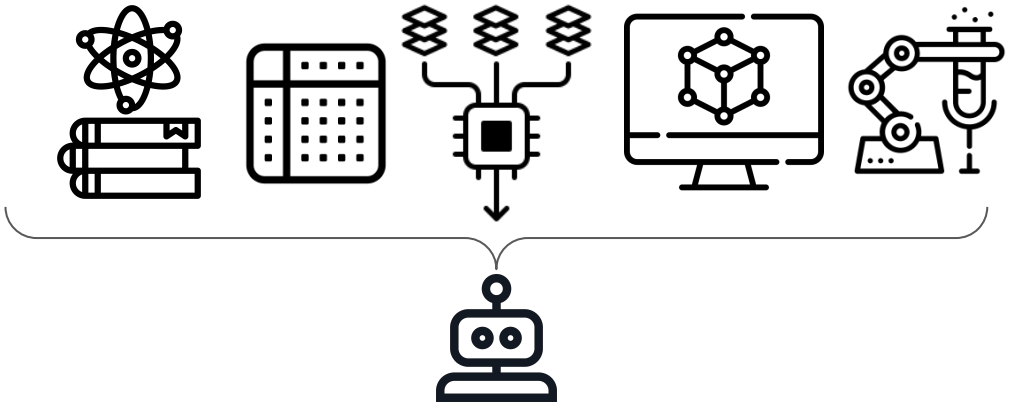
NEWS | 29 November 2023

Google AI and robots join forces to build new materials

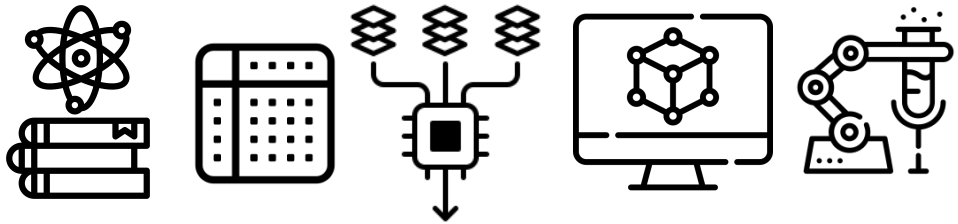
Tool from Google DeepMind predicts nearly 400,000 stable substances, and an autonomous system learns to make them in the lab.



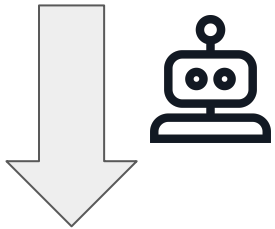
Literature, datasets, models, simulators, robotic labs



Literature, datasets, models, simulators, robotic labs

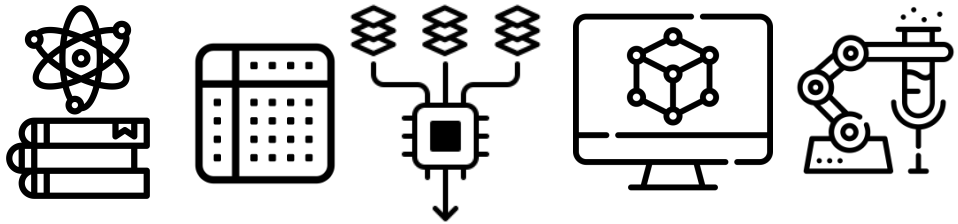


Come up with
hypotheses
(what to explore)

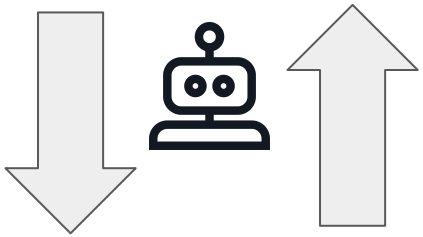




Literature, datasets, models, simulators, robotic labs



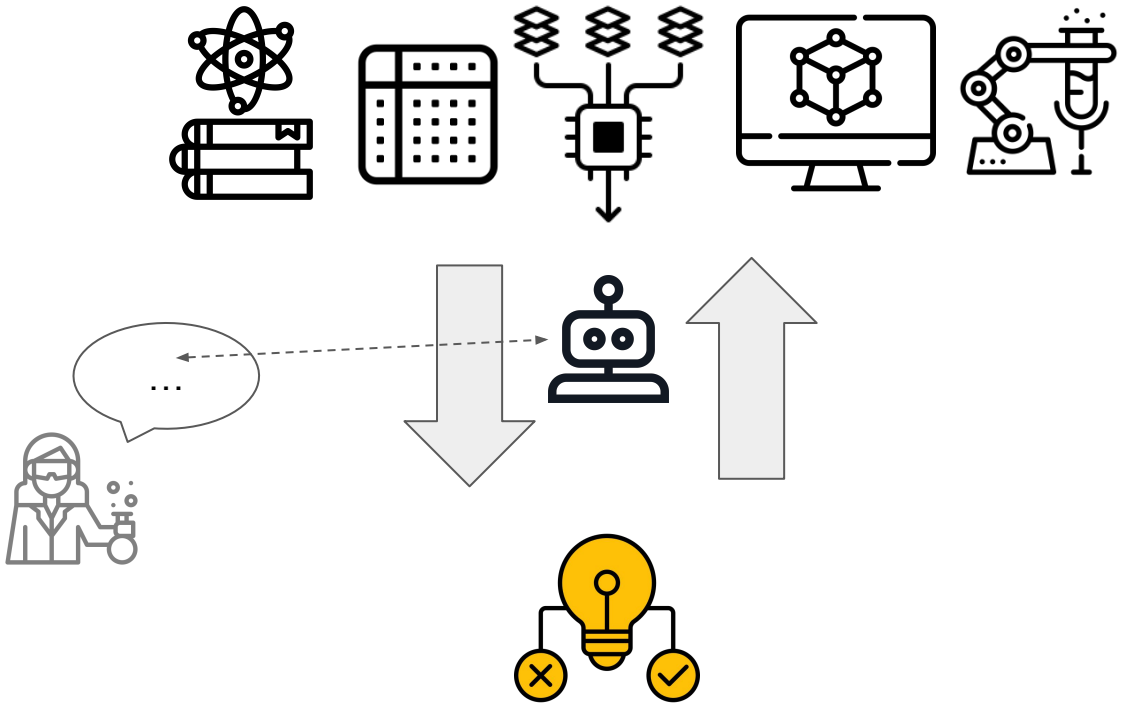
Come up with
hypotheses
(what to explore)



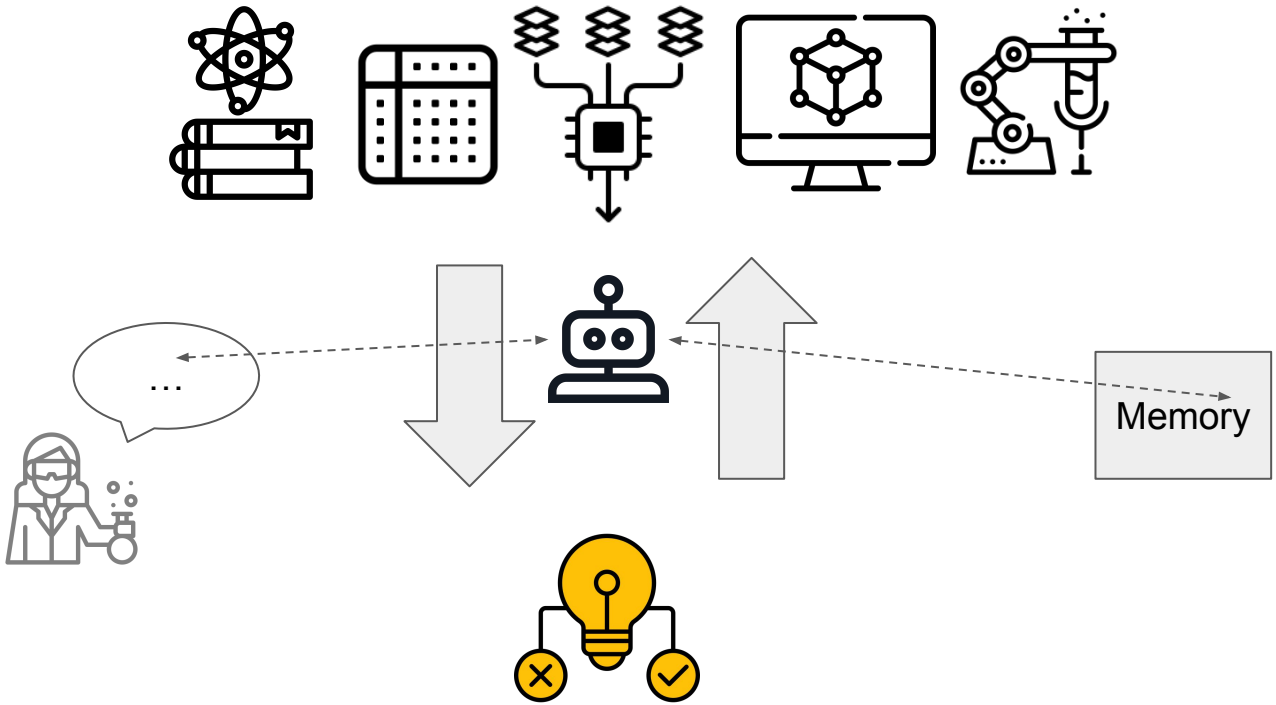
Closed-loop evaluation
(run the experiments, validate)



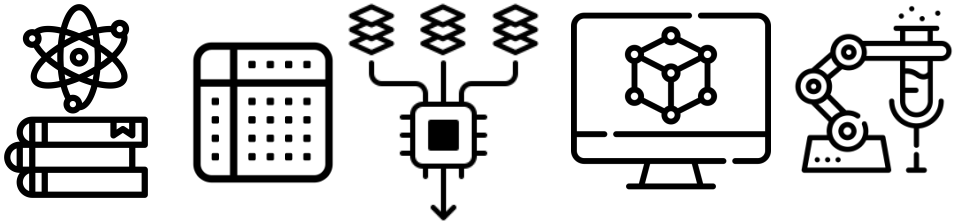
Literature, datasets, models, simulators, robotic labs

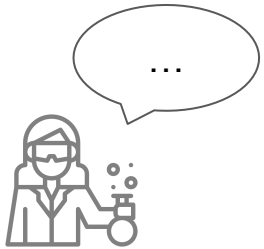


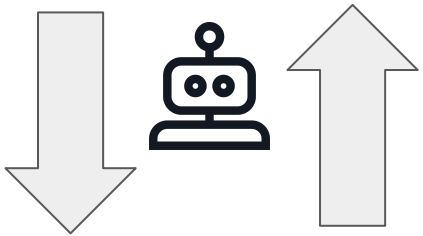
Literature, datasets, models, simulators, robotic labs



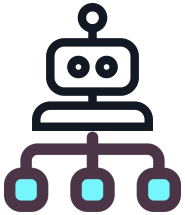
Literature, datasets, models, simulators, robotic labs











Hierarchical multi-agent setup: Ideation agents, critique agents, experiment planning, execution…


Research
Tasks



Attention to areas of interest
(How do we keep track?)

Problem identification & prioritization
(How do we select what to work on?)



Forming directions
(How do we generate solutions?)





Learning, understanding, critiquing

Experimentation, analysis

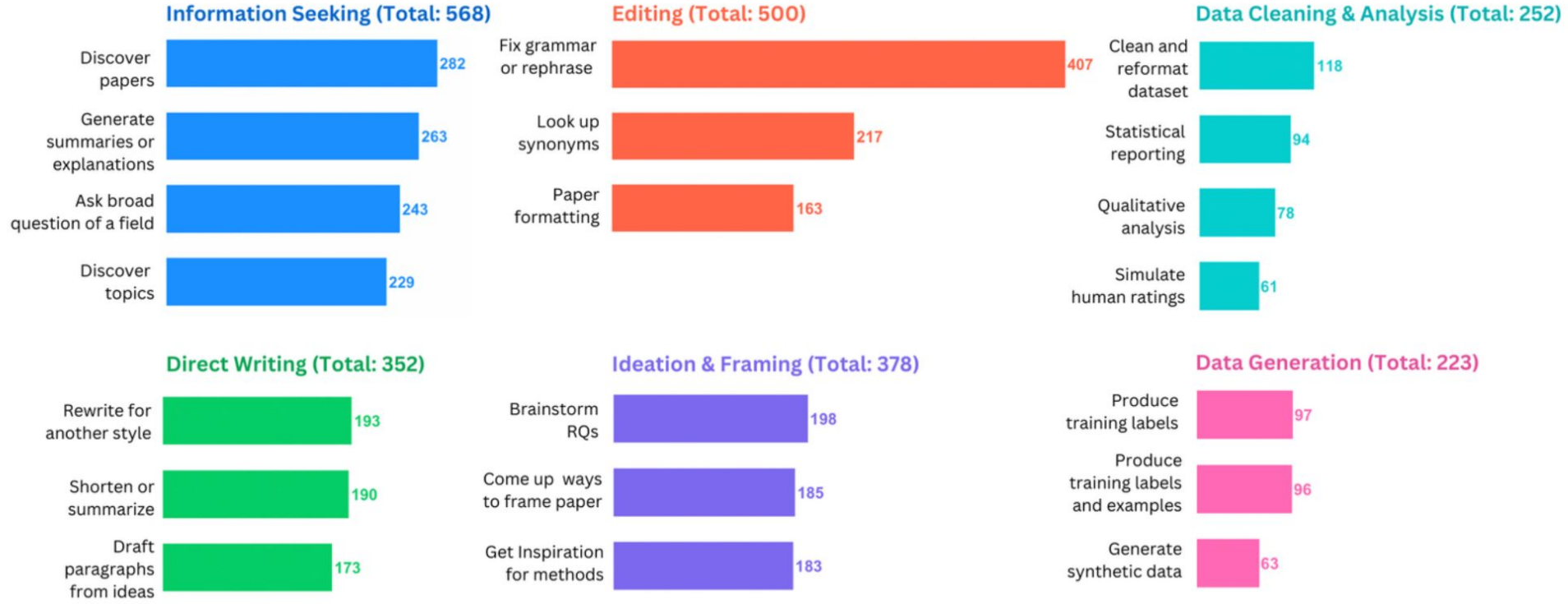




Communication, collaboration

How do you / would you

use LLMs/AI for **your research**?



LLMs as Research Tools: A Large Scale Survey of Researchers' Usage and Perceptions

sakana.ai

The AI Scientist Generates its First Peer-Reviewed Scientific Publication

March 12, 2025

NEWS | 29 November 2023

Google AI and robots join forces to build new materials

Tool from Google DeepMind predicts nearly 400,000 stable substances, and an autonomous system learns to make them in the lab.



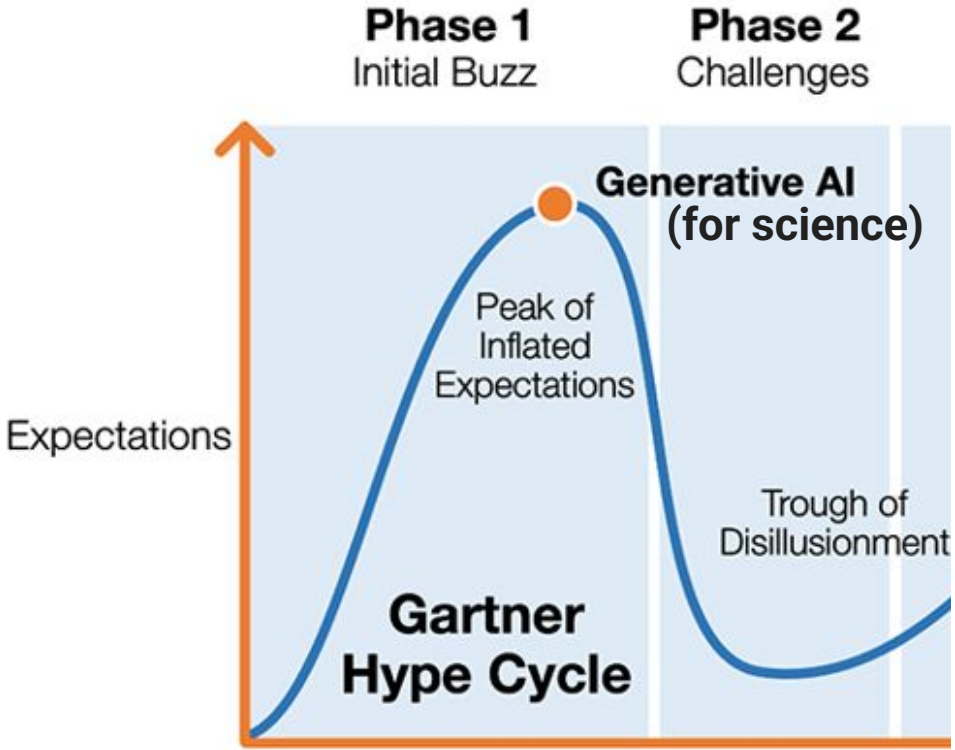
DeepMind AI outdoes human mathematicians on unsolved problem

Large language model improves on efforts to solve combinatorics problems

Google Co-Scientist AI cracks superbug problem in two days!

This AI just figured out geometry — is this a step towards artificial reasoning?

How 'AlphaGeometry' solves Mathematical Olympiad-level problems


Phase 1
Initial Buzz
Phase 2
Challenges
Generative AI
(for science)
Peak of
Inflated
Expectations
Expectations
Trough of
Disillusionment
Gartner
Hype Cycle



Science of Science @MishaTeplitskiy · Dec 15, 2023

Another day, another case of "AI accelerates scientific discovery!!!" that we need to pump the brakes on
nature.com/articles/d4158...

Google Co-Scientist AI cracks superbug problem in two days!

Google Co-Scientist AI cracks superbug problem in two days! – because it had been fed the team's previous paper with the answer in it



DAVID GERARD / 22 FEBRUARY 2025 / SCIENCE

Google AI and robots join forces to build new materials



Robert Palgrave
@Robert_Palgrave

This exciting paper shows AI design of materials, robotic synthesis. 10s of new compounds in 17 days.

But did they?

Google AI and robots join forces to build new materials



Robert Palgrave @Robert_Palgrave · Jan 8

The headline conclusions:

It is likely that no new materials were discovered.

Most of the materials produced were misidentified, and the rest were already known

This AI just figured out geometry — is this a step towards artificial reasoning?

How 'AlphaGeometry' solves Mathematical Olympiad-level problems

DeepMind AI outdoes human mathematicians on unsolved problem

Large language model improves on efforts to solve combinatorics problems

tasks[26–29]. When producing full natural-language proofs on IMO-AG-30, however, GPT-4 has a success rate of 0%, often making syntactic and semantic errors throughout its outputs, showing little understanding of geometry knowledge and of the problem statements itself. Note that

1. Trained on **~100 million** synthetically generated proofs (narrow, specific kind of proof)

2. Heavily relies on a **non-LLM symbolic/logic engine (exists for decades).**



THE LOGIC THEORY MACHINE
A COMPLEX INFORMATION PROCESSING SYSTEM

by

Allen Newell and Herbert A. Simon

P-868

June 15, 1956

Human proof: elegant, **creative**, short.

Original problem statement:

Let AH_1, BH_2, and CH_3 be the altitudes of a triangle ABC. The incircle W of triangle ABC touches the sides BC, CA and AB at T_1, T_2 and T_3, respectively. Consider the symmetric images of the lines H_1H_2, H_2H_3, and H_3H_1 with respect to the lines T_1T_2, T_2T_3, and T_3T_1. Prove that these images form a triangle whose vertices lie on W.

Human proof (from Evan Chen's collection)

We use complex numbers with ω the unit circle. Let $T_1 = a$, $T_2 = b$, $T_3 = c$. The main content of the problem is to show that the triangle in question has vertices ab/c, bc/a, ca/b (which is evident from a good diagram).

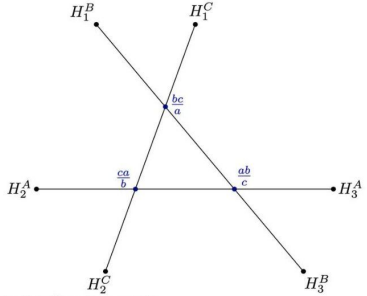
Since $A = \frac{2bc}{b+c}$, we have

$$H_1 = \frac{1}{2}\left(\frac{2bc}{b+c} + a + a - a^2 \cdot \frac{2}{b+c}\right) = \frac{ab+bc+ca-a^2}{b+c}.$$

The reflection of H_1 over $\overline{T_1T_2}$ is

$$H_1^C = a + b - ab\overline{H_1} = a + b - b \cdot \frac{ac+ab+a^2-bc}{a(b+c)}$$
$$= \frac{a(a+b)(b+c) - b(a^2+ab+ac-bc)}{a(b+c)} = \frac{c(a^2+b^2)}{a(b+c)}.$$

Now, we claim that H_1^C lies on the chord joining $\frac{ca}{b}$ and $\frac{cb}{a}$; by symmetry so will H_2^C and this will imply the problem (it means that the desired triangle has vertices ab/c, bc/a, ca/b). A cartoon of this is shown below.



To see this, it suffices to compute

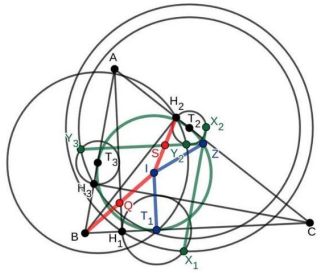
$$H_1^C + \left(\frac{ca}{b}\right)\left(\frac{cb}{a}\right)\overline{H_1^C} = \frac{c(a^2+b^2)}{a(b+c)} + c^2 \frac{\frac{1}{c} \cdot \frac{a^2+b^2}{a^2b^2}}{\frac{1}{a}\left(\frac{b+c}{bc}\right)}$$
$$= \frac{c(a^2+b^2)}{a(b+c)} + \frac{c(a^2+b^2)}{abc^{-1}(b+c)}$$
$$= \frac{c(a^2+b^2)}{a(b+c)}\left(\frac{b+c}{b}\right)$$
$$= \frac{c(a^2+b^2)}{ab} = \frac{ca}{b} + \frac{cb}{a}$$

as desired.

Adapted problem statement:

Let ABC be a triangle. Define point I such that AI is the bisector of ∠ BAC and CI is the bisector of ∠ ACB. Define point T_1 as the foot of I on line BC. Define point T_2 as the foot of I on line AC. Define point T_3 as the foot of I on line AB. Define point H_1 as the foot of A on line BC. Define point H_2 as the foot of B on line AC. Define point H_3 as the foot of C on line AB. Define point X_1 as the intersection of circles (T_1,H_1) and (T_2,H_1). Define point X_2 as the intersection of circles (T_1,H_2) and (T_2,H_2). Define point Y_2 as the intersection of circles (T_2,H_2) and (T_3,H_2). Define point Y_3 as the intersection of circles (T_2,H_3) and (T_3,H_3). Define point Z as the intersection of lines X_1X_2 and Y_2Y_3. Prove that $T_1I = IZ$.

AlphaGeometry proof (90% proof lines omitted)



Construct point Q as the midpoint of BI.

Construct point S as the midpoint of IH_2.

Step 1. $T_2H_2 = T_2Y_2 \Rightarrow \angle T_2H_2Y_2 = \angle H_2Y_2T_2$.

Step 2. $T_2H_3 = T_2Y_3$ and $T_3H_3 = T_3Y_3 \Rightarrow T_2T_3 \perp H_3Y_3$.

Step 3. $T_2H_3 = T_2Y_3$ and $T_3H_3 = T_3Y_3 \Rightarrow \angle T_2H_3T_3 = \angle T_3Y_3T_2$.

Step 4. $T_2H_2 = T_2Y_2$ and $T_3H_2 = T_3Y_2 \Rightarrow T_2T_3 \perp H_2Y_2$.

Step 5. A, C, T_2 are collinear, A, C, H_2 are collinear, $\angle T_2H_2Y_2 = \angle H_2Y_2T_2$, $T_2T_3 \perp H_2Y_2$ and $T_2T_3 \perp H_3Y_3 \Rightarrow$ ∠AC, $H_3Y_3 = \angle H_3Y_3$, T_2Y_2.

… [90% omitted]…

Step 106. ∠ IT_2Z = ∠ QH_2Z and $T_2I:T_2Z = H_2Q:H_2Z \Rightarrow$ $\angle(T_2I, H_2Q) = \angle IZQ$ and $T_2I:H_2Q = ZI:ZQ$.

Step 107. $\angle(T_2I, H_2Q) = \angle IZQ$, QS is the bisector of $\angle T_2QH_2$, $BH_2 \parallel QS$, $AC \perp BH_2$ and $AC \perp T_2I \Rightarrow \angle QT_2I = \angle IZQ$.

Step 108. $T_2Q = H_2Q$ and $T_2I:H_2Q = ZI:ZQ \Rightarrow T_2I:T_2Q = ZI:ZQ$.

Step 109. $\angle QT_2I = \angle IZQ$ and $T_2I:T_2Q = ZI:ZQ \Rightarrow T_2I = IZ$.

Step 110. $T_1I = T_2I$ and $T_2I = IZ \Rightarrow T_1I = IZ$

"AI" proof:

> 100 technical deductive steps

only 2 steps come from the LLM; The rest from the symbolic engine (with which LLM was trained).

My own experiments on scientific idea generation :

LLMs **generate directions that lack novelty/creativity**

SciMON: Scientific Inspiration Machines Optimized for Novelty, ACL 2024

CAN LLMS GENERATE NOVEL RESEARCH IDEAS?
A LARGE-SCALE HUMAN STUDY WITH 100+ NLP RESEARCHERS

Chenglei Si, Diyi Yang, Tatsunori Hashimoto
Stanford University
{clsi,diyiy,thashim}@stanford.edu

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All That Glitters is Not Novel: Plagiarism in AI Generated Research

Tarun Gupta
Indian Institute of Science
Bengaluru, KA, India
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Danish Pruthi
Indian Institute of Science
Bengaluru, KA, India
danishp@iisc.ac.in

What capabilities do you think AI scientists need?

Literature Search and Understanding

Ideation - generating “good” directions

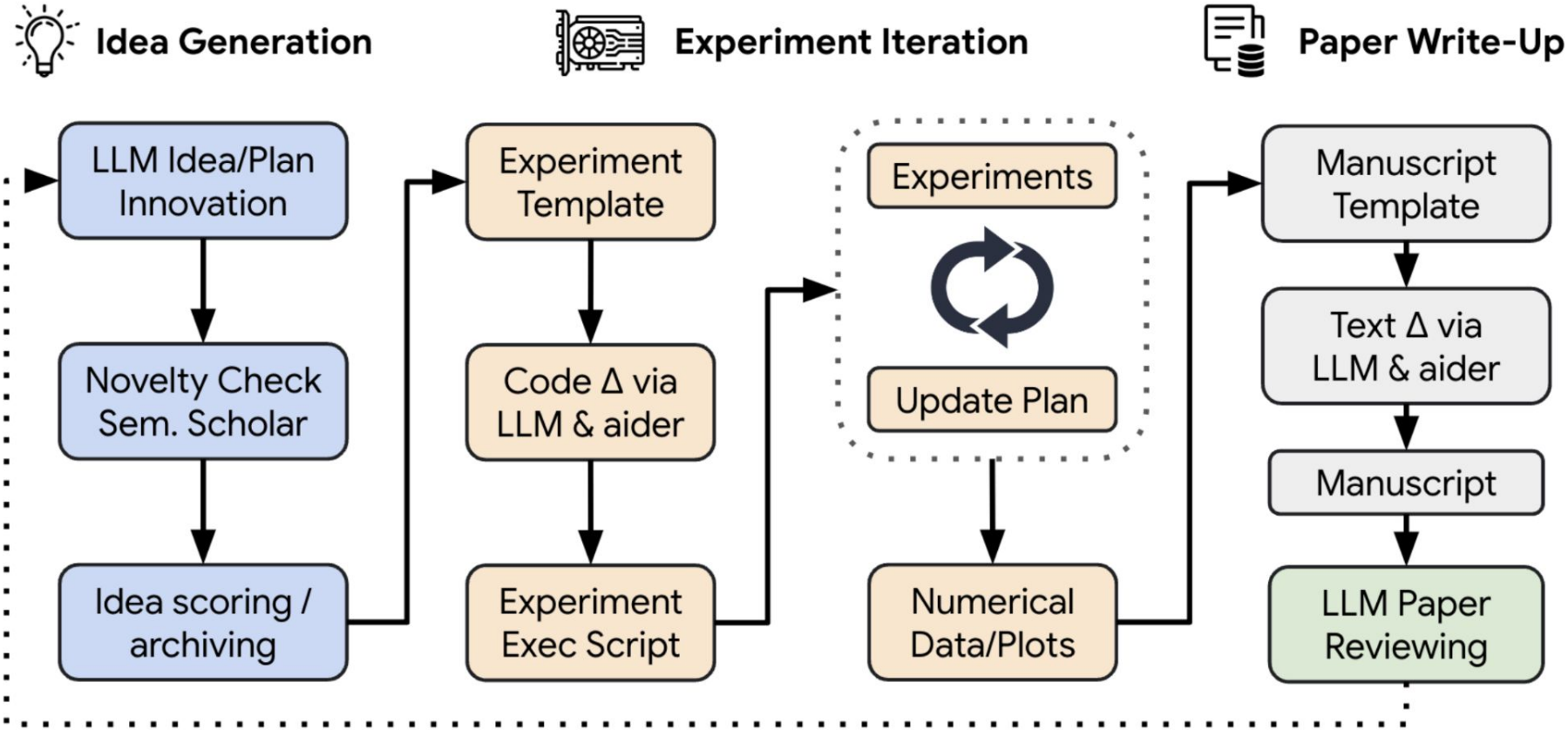
Forming and executing a research plan

Data analysis

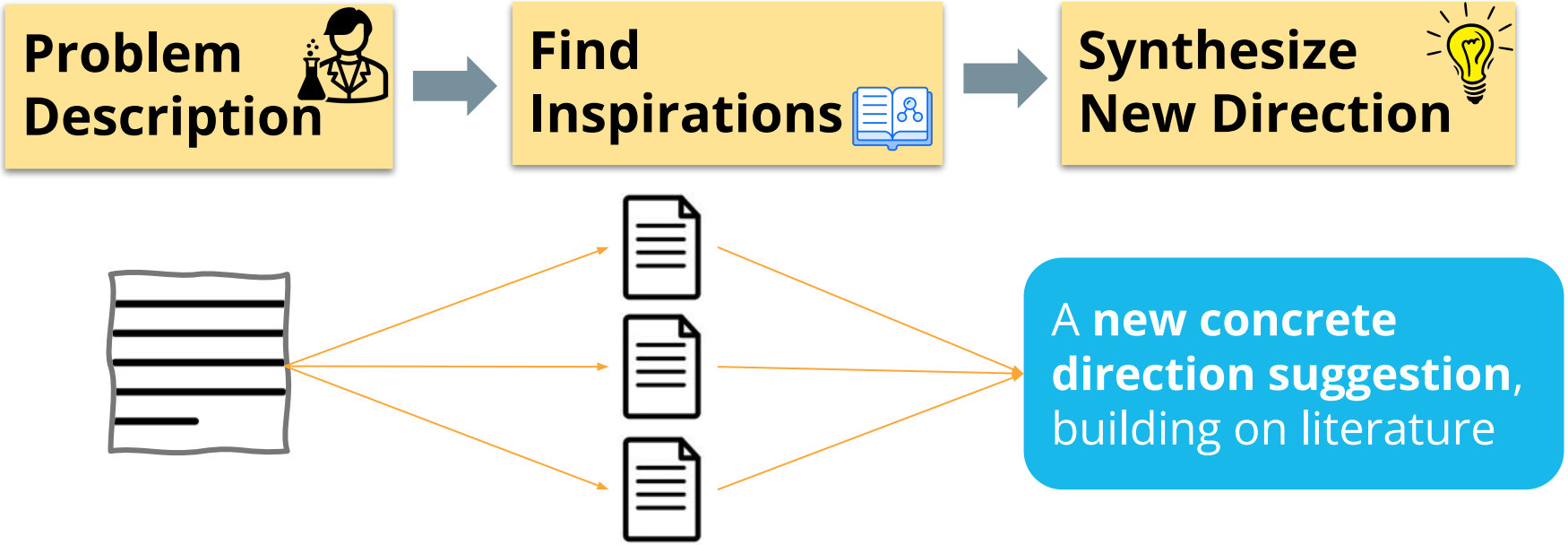
Critiquing quality, novelty

Drawing meaningful conclusions, insights

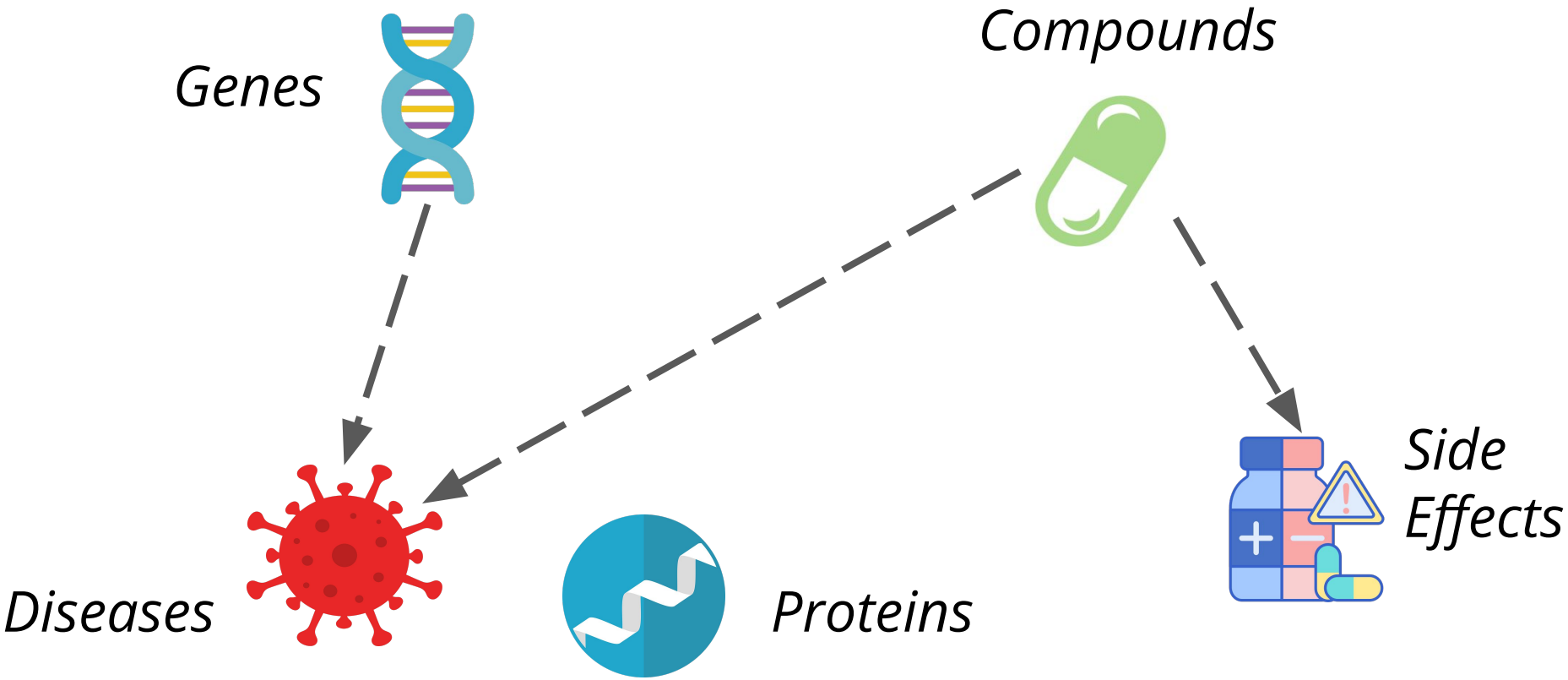
?


Idea Generation
Experiment Iteration
Paper Write-Up
LLM Idea/Plan Innovation
Novelty Check Sem. Scholar
Idea scoring / archiving
Experiment Template
Code Δ via LLM & aider
Experiment Exec Script
Experiments
Update Plan
Numerical Data/Plots
Manuscript Template
Text Δ via LLM & aider
Manuscript
LLM Paper Reviewing

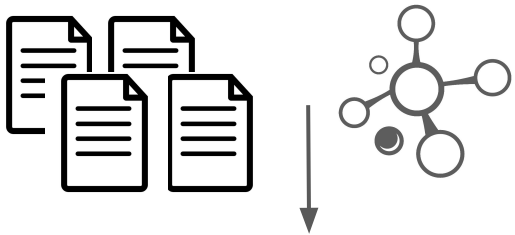
Ideation - generating “good” directions



Predicting Links in Knowledge Graphs



Ingest scientific corpora, resources

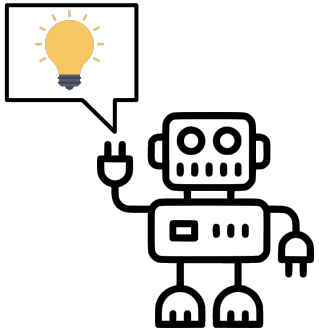


Extraction, mining, inference



Identify **blindspots, gaps, opportunities** in the "space of ideas"

Hypothesize and **prioritize** novel and feasible **directions**





Google AI co-scientist Trusted Tester Program

Critiquing quality, novelty

Your task now is to draft a high-quality review outline for a top-tier Machine Learning (ML) conference submission:

Parsed PDF

Your task:
Compose a high-quality peer review of an ML paper submitted to a top-tier ML conference on OpenReview.

Start by "Review outline:".
And then:

"1. Significance and novelty"

"2. Potential reasons for acceptance"

"3. Potential reasons for rejection", List multiple key reasons.

"4. Suggestions for improvement", List multiple key suggestions.

Be thoughtful and constructive. Write outlines only.



Review outline:

1. Significance and novelty

1. Introduction of the modality gap...
2. ...

2. Potential reasons for acceptance

1. Theoretical and empirical analyses...
2. ...
3. ...

3. Potential reasons for rejection

1. Lack of a proposed method...
2. ...
3. ...

4. Suggestions for improvement

1. Expand the range of experiments...
2. ...
3. ...

tomh Today, 6:17 p.m.

review this paper: https://www.nber.org/system/files/working_papers/w33020/w33020.pdf

nora Today, 6:17 p.m.

The review for the paper provided via the URL "https://www.nber.org/system/files/working_papers/w33020/w33020.pdf" is currently in progress and is expected to take about 10 minutes. I will notify you as soon as the review is completed.

nora Today, 6:22 p.m.

The review for the paper provided via the URL "https://www.nber.org/system/files/working_papers/w33020/w33020.pdf" has been completed. You can View report › .

If you have any follow-up questions or need further clarification on the review, feel free to ask!

is there another way I can assist you with this paper?

U **tomh** Today, 6:17 p.m.

review this paper: https://www.nber.org/system/files/working_papers/w33020/w33020.pdf

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nora Today, 6:22 p.m.

The review for the paper provided via the URL "https://www.nber.org/system/files/working_papers/w3302

Let's do some research

Suggesting a Control Group

Include a control group to better demonstrate the effectiveness of the teaching methods.

Clarify DA Mechanism Definitions

Clarity of Definitions: The paper should provide clearer operational definitions of 'traditional' and 'simplified' descriptions of the Deferred Acceptance mechanism. The terms 'Traditional DA Mechanics (Trad-DA)' and 'Menu DA Mechanics (Menu-DA)' are mentioned, but it is not explicitly stated how these relate to the traditional and simplified categories. Clear definitions are crucial for replicability and understanding of the study.

Strengthening Assumption Justifications

Justification of Assumptions: While the paper assumes that simplified descriptions inherently lead to better understanding and strategyproof behavior, it would benefit from a more detailed justification of this assumption. Including a review of literature that supports this

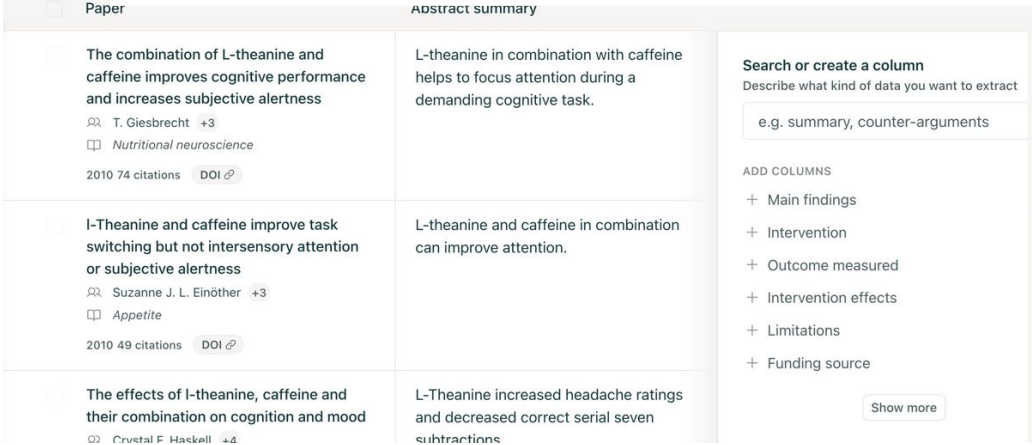
(No)

Is LLM a Reliable Reviewer? A Comprehensive Evaluation of LLM on Automatic Paper Reviewing Tasks

Ruiyang Zhou, Lu Chen, Kai Yu

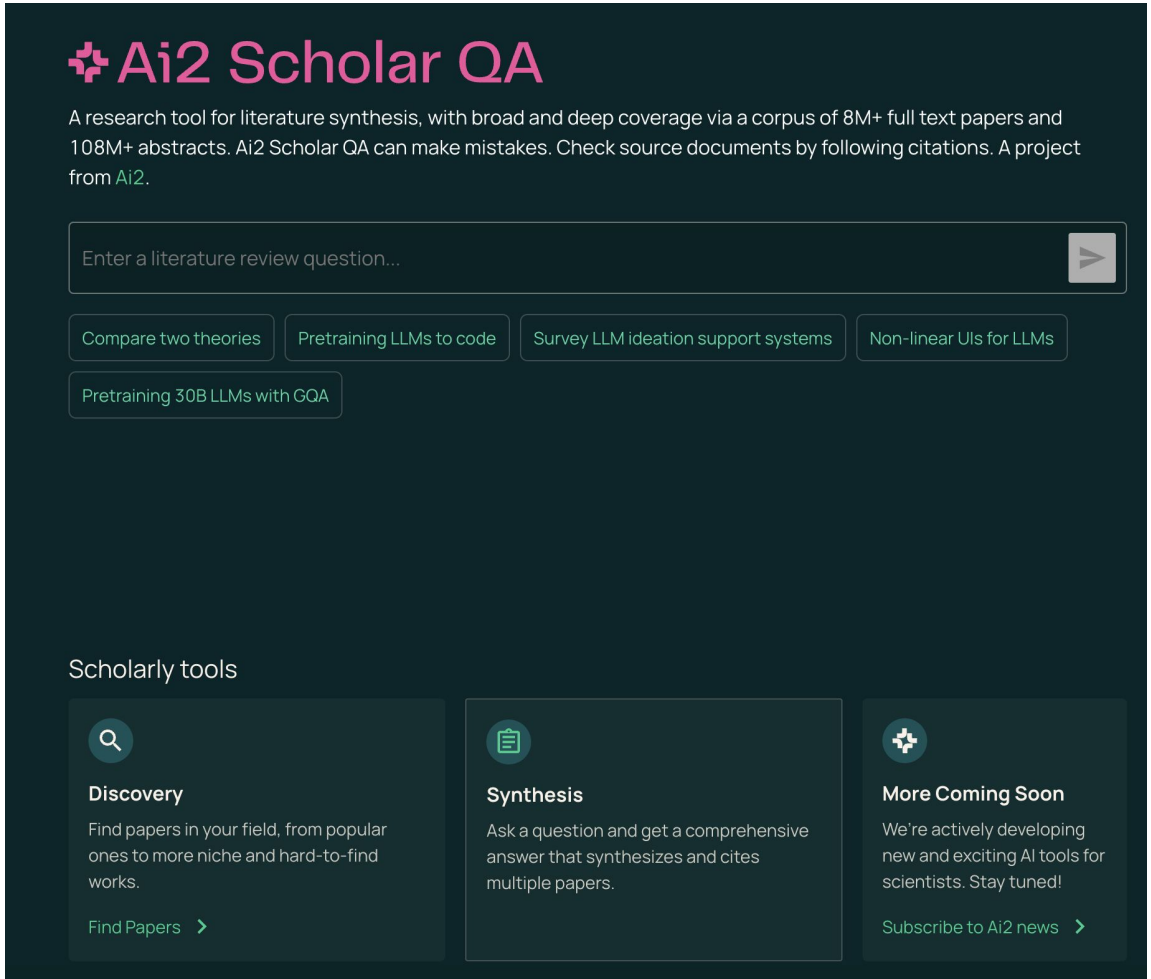
Literature Search and Understanding

	Dataset size	Annotation method	Intended Application	Evaluation Metric
Paper 1	1,200 video sequences	Subjectively annotated	Objective VQA method development	Subjective Mean Opinion Score
Paper 2	585 videos	Subjective video quality scores via crowdsourcing	NR video quality prediction advancement	Subjective video quality scores
Paper 3	153,841 videos	Coarsely annotated set with five quality ratings each	Deep-learning VQA model training	Spearman rank-order correlation coefficient
Paper 4	1 million YouTube videos	N/A	Large-scale video classification and action recognition	Performance improvements over baselines

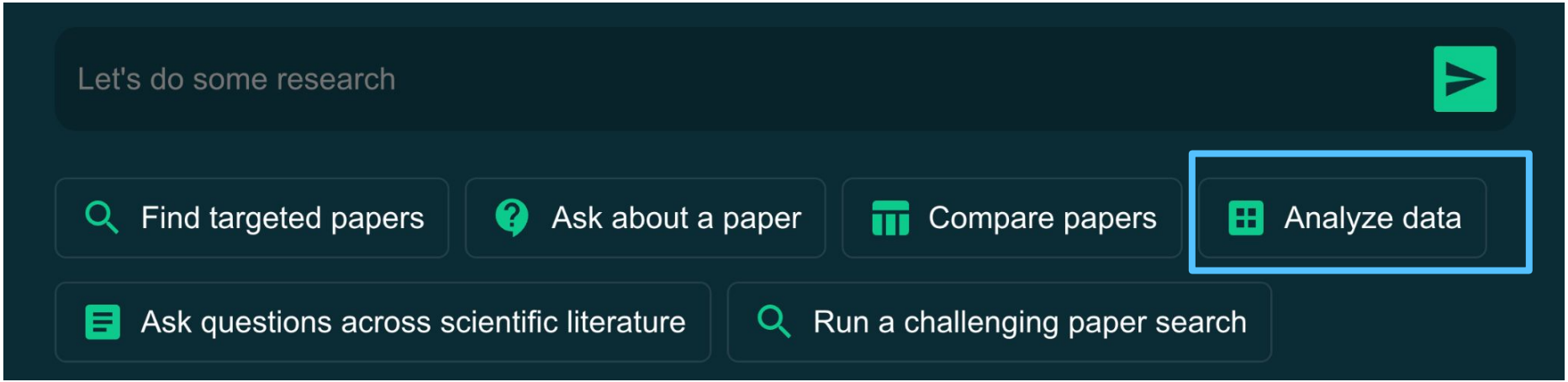


https://elicit.com/

Literature Search and Understanding



Data analysis



Can you analyze the relation between time preference and BMI in this dataset: https://raw.githubusercontent.com/allenai/discoverybench/main/discoverybench/real/train/nls_bmi/nls_bmi_processed.csv

Find targeted papers | Ask about a paper | Compare papers | Analyze data

Ask questions across scientific literature | Run a challenging paper search

Let's move on to the data exploration step to understand the relationship between time preference and BMI. We'll start by examining the distribution of these variables and then analyze their relationship.

```python
import matplotlib.pyplot as plt
import seaborn as sns

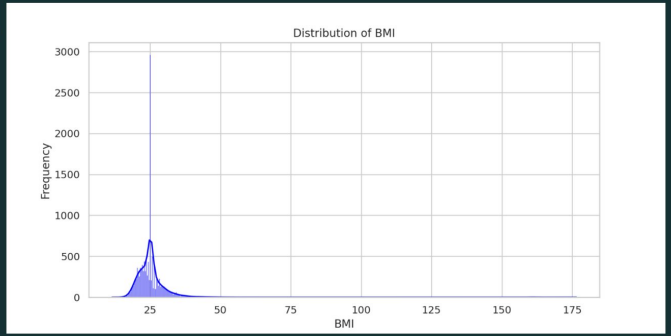
# Set the style of seaborn
sns.set(style="whitegrid")

# Plot the distribution of BMI
plt.figure(figsize=(10, 5))
sns.histplot(df['BMI'], kde=True, color='blue')
plt.title('Distribution of BMI')
plt.xlabel('BMI')
plt.ylabel('Frequency')
plt.show()

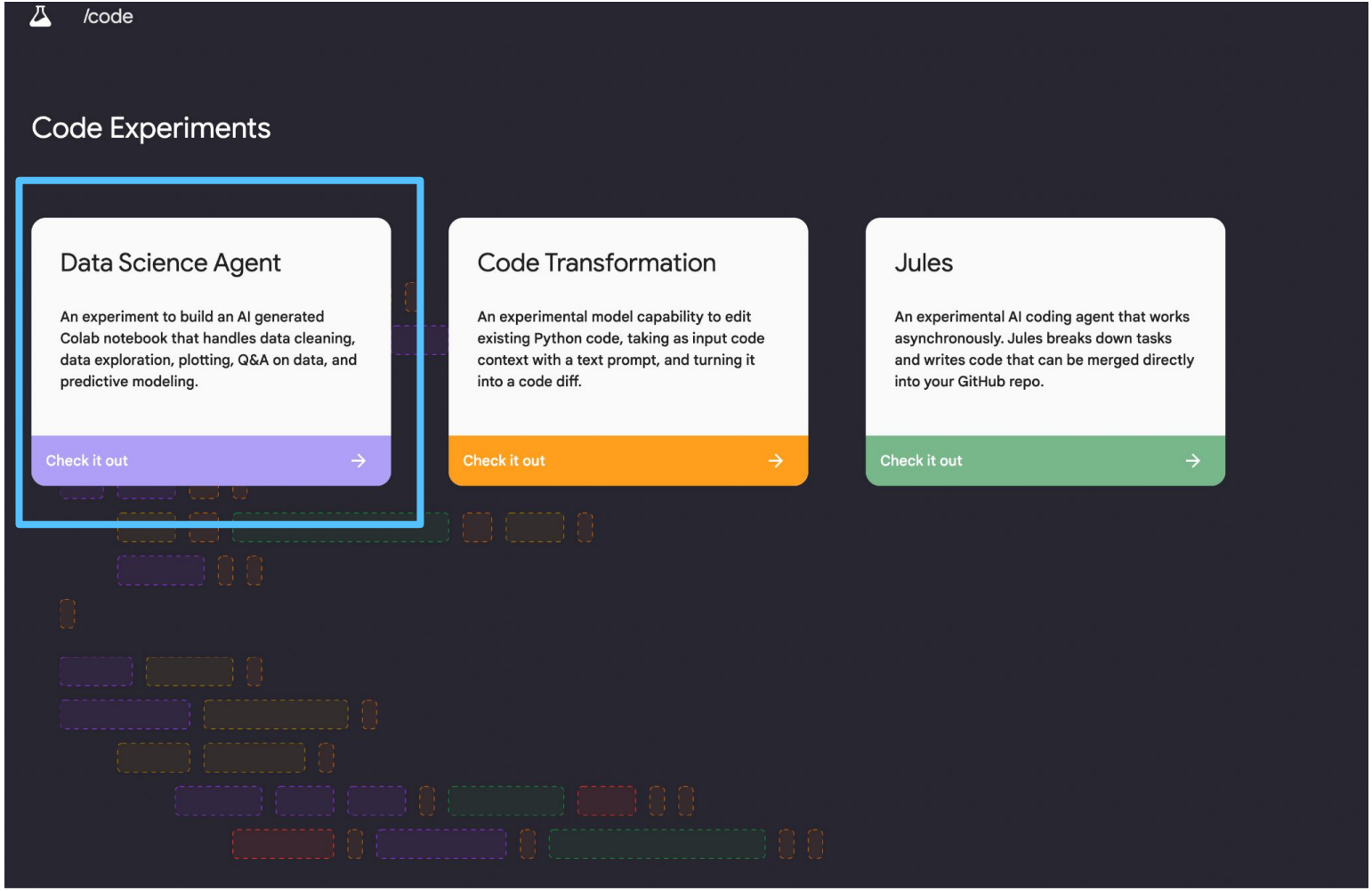
# Plot the distribution of DISSAVED and SAMESAVE
plt.figure(figsize=(10, 5))
sns.countplot(data=df, x='DISSAVED', palette='pastel')
plt.title('Distribution of DISSAVED')
plt.xlabel('DISSAVED')
plt.ylabel('Count')
plt.show()

plt.figure(figsize=(10, 5))
sns.countplot(data=df, x='SAMESAVE', palette='pastel')
plt.title('Distribution of SAMESAVE')
plt.xlabel('SAMESAVE')
plt.ylabel('Count')
plt.show()
```





https://labs.google.com/code/





Researcher Inner World



Researcher Inner World



State of Knowledge

Researcher Inner World



State of Knowledge

Preferences

Researcher Inner World



State of Knowledge

Preferences

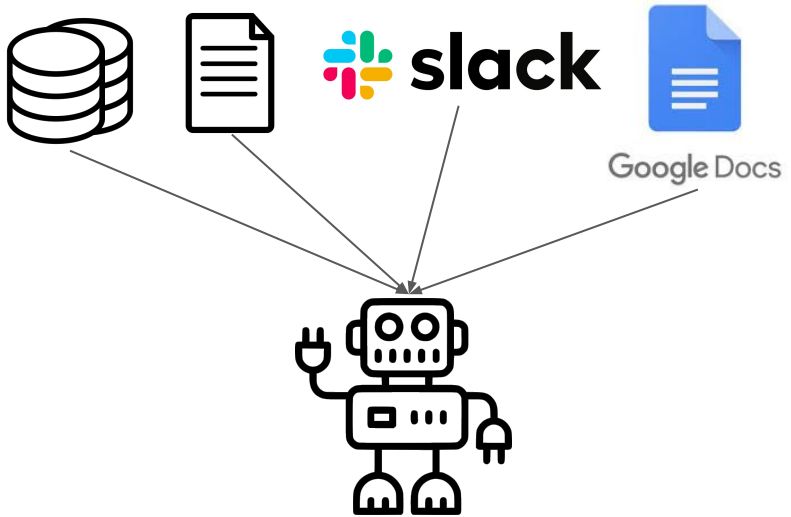
Key Tasks: selecting problems, forming directions…

Researcher Inner World

State of Knowledge
Preferences

Key Tasks: selecting problems, forming directions…

research assistance systems with access to users' documents, data, experiments, communication channels…

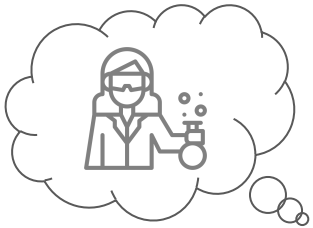


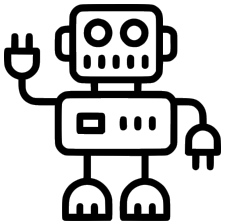
Researcher Inner World

State of Knowledge
Preferences

Key Tasks: selecting problems, forming directions…

Using this data, systems make **inferences on users' goals, difficulties, knowledge**



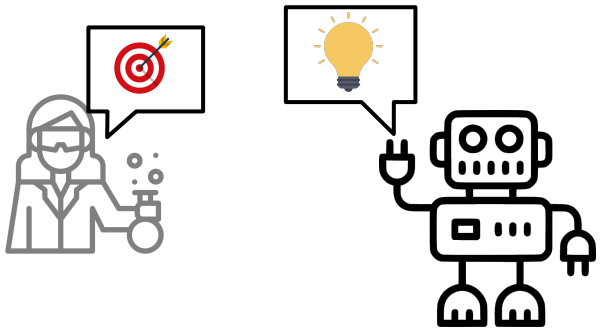


Researcher Inner World

State of Knowledge
Preferences

Key Tasks: selecting problems, forming directions...

Engage in conversational interaction to suggest **solutions, hypotheses and experiments** and execute them



Scientific Knowledge: Challenges



Large-scale, diverse

Rapidly evolving

Deeply technical

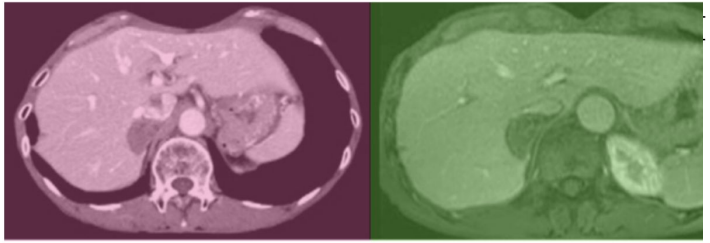
$$\frac{\mathbf{z}_i^{(k+1)} - \mathbf{z}_i^{(k)}}{\tau} = \sum_{j:(i,j)\in\mathcal{E}(\mathbf{U}^{(k)})} a\left(\mathbf{z}_i^{(k)}, \mathbf{z}_j^{(k)}\right)(\mathbf{z}_j^{(k)} - \mathbf{z}_i^{(k)}). \tag{9}$$

Here k denotes the discrete time index (iteration) and τ is the time step (discretisation parameter). Rewriting (9) compactly in matrix-vector form with $\tau = 1$ leads to the *explicit Euler scheme*:

$$\mathbf{Z}^{(k+1)} = (\mathbf{A}^{(k)} - \mathbf{I})\mathbf{Z}^{(k)} = \mathbf{Q}^{(k)}\mathbf{Z}^{(k)}, \tag{10}$$

where $a_{ij}^{(k)} = a(\mathbf{z}_i^{(k)}, \mathbf{z}_j^{(k)})$ and the matrix $\mathbf{Q}^{(k)}$ (diffusion operator) is given by

$$q_{ij}^{(k)} = \begin{cases} 1 - \tau \sum_{l:(i,l)\in\mathcal{E}} a_{il}^{(k)} & i = j \\ \tau a_{ij}^{(k)} & (i,j) \in \mathcal{E}(\mathbf{U}^{(k)}) \\ 0 & \text{otherwise} \end{cases}$$



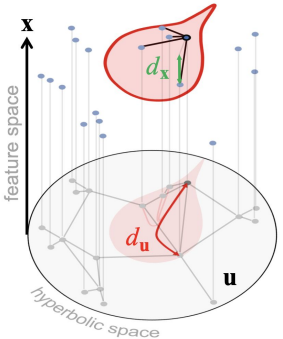
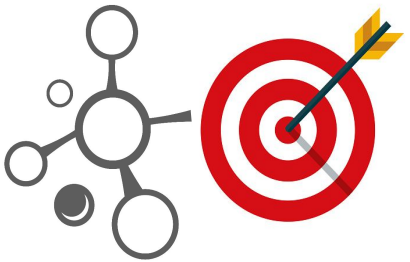


Figure 2: Graph Beltrami flow with

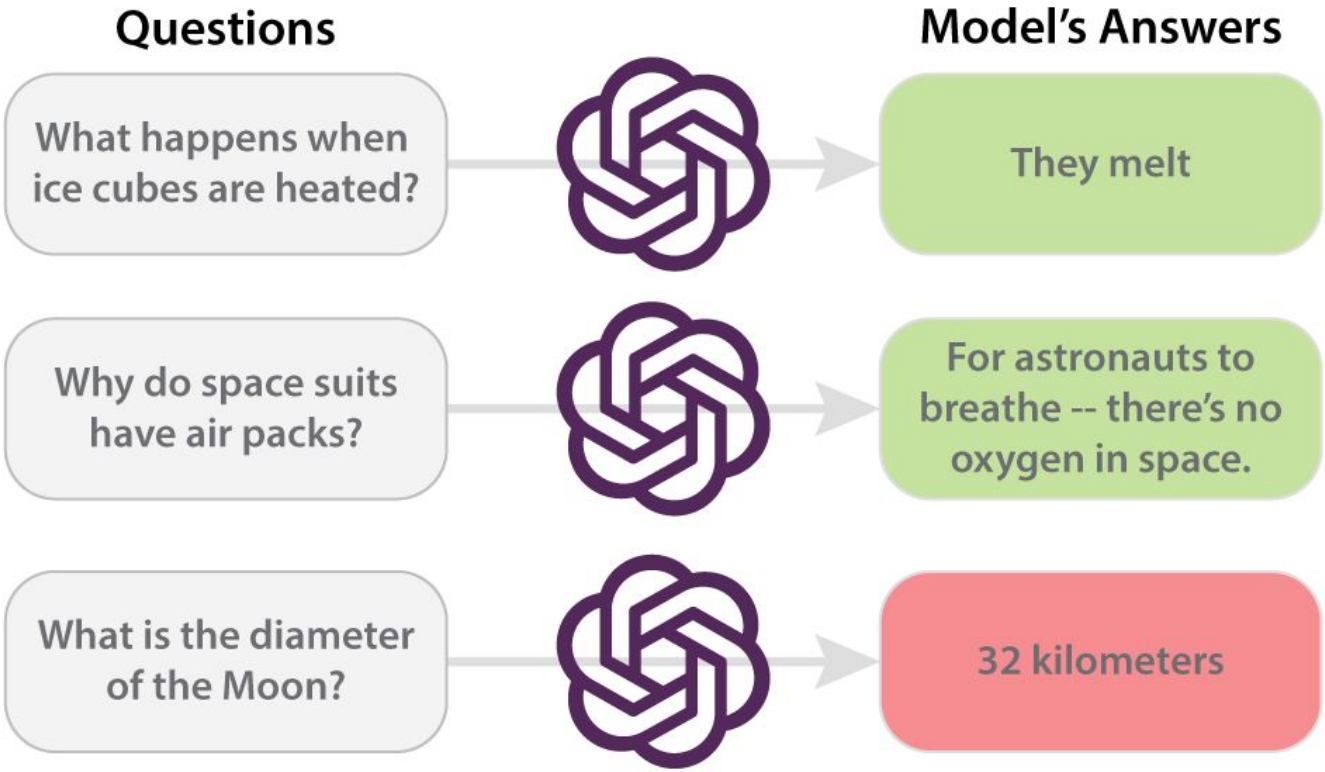
Prediction and prioritization

- How to identify promising areas in large “spaces of ideas”

- How to **prioritize directions** in these spaces:
  - Novel
  - Plausible
  - Valuable



How do we normally evaluate machine learning models?



Model Performance = 2/3 = 66%

How do we normally evaluate machine learning models?

Questions

Challenge

In scientific discovery, we're trying to discover new knowledge

So, we don't have the answers!

How do we test if a model can make new discoveries?

Model's Answers

They melt

For astronauts to breathe -- there's no oxygen in space.

32 kilometers

Model Performance = 2/3 = 66%



Forms of Discovery

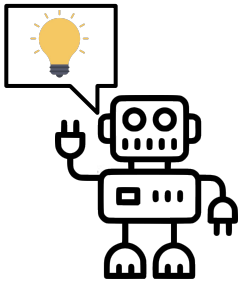
Science is not a single activity. Few scientists carry on, even in the course of a whole career, the entire range of activities that belong to science. It is easy to identify at least seven different classes of activities that must be regarded as forms of scientific research:

1. **Finding patterns in data (that is, descriptive theories)**
2. **Finding good problems for investigation**
3. **Finding new data (observing and experimenting, including designing experimental strategies)**
4. **Inventing instruments**
5. **Inventing representations for classes of problems**
6. **Inventing new theoretical concepts**
7. **Finding explanations (explanatary theories) for data or descriptive theories.**

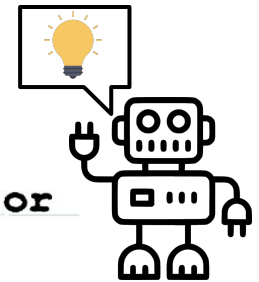


Forms of Discovery

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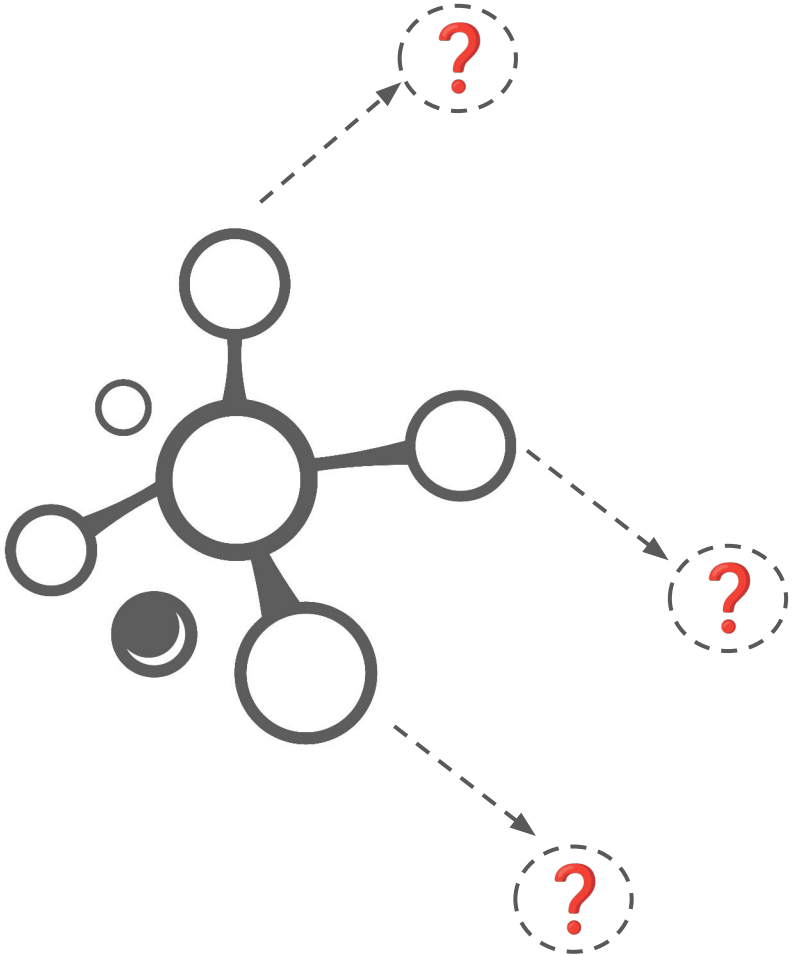


1. Finding patterns in data (that is, descriptive theories)
2. Finding good problems for investigation
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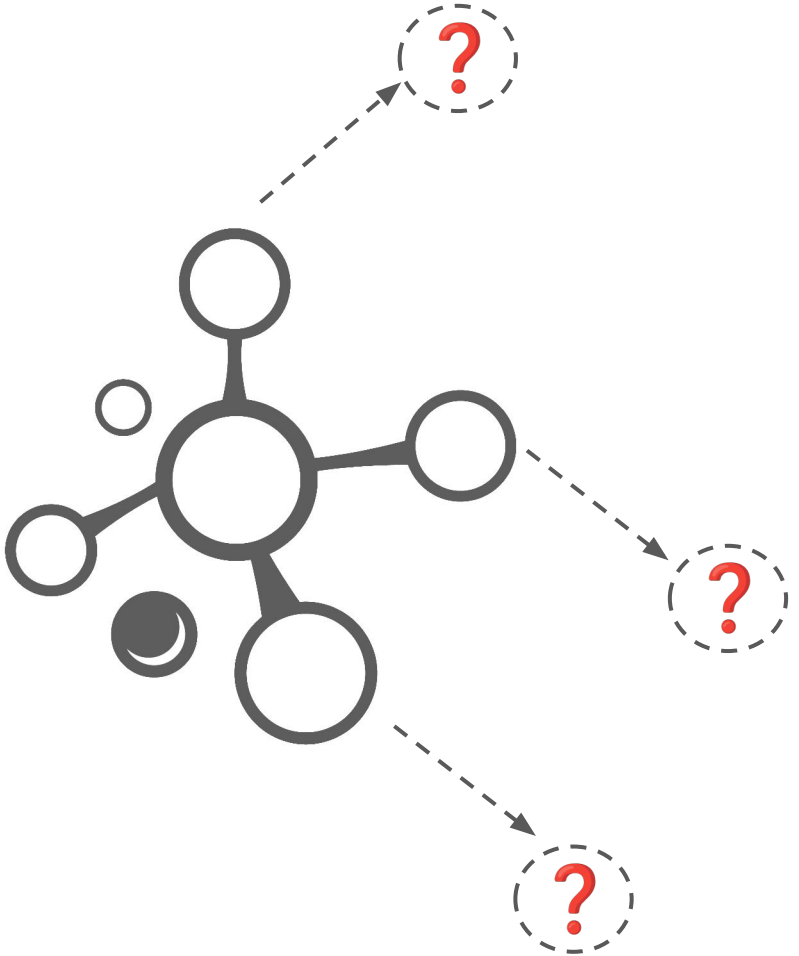
Science can evolve along countless **possible trajectories** --

affected by social determinants, human lenses on reality.



Will current and future AI merely **accelerate** our progress along “human trajectories”,

or steer us along **fundamentally different** paths?



What this course is

- Assumes a lot of prior knowledge in NLP, LLMs.
- Very much centered around project-based learning:
  - Lightweight and just a couple lectures, quick introduction to some core ideas, showing research examples.

- יתקיימו בשבועות הראשונים לסמסטר הרצאות רגילות.

- לאחר מכן תהיה הפסקה של מספר שבועות משיעורים פרונטליים, במקום זאת בזמן המיועד לשיעור כל זוג יקבע פגישה או שתיים איתי למחשבה משותפת על נושא כללי לפרויקט. במהלך שבועות אלה, אתם אמורים לנצל את הזמן לעיין במאמרים שאשתף, להתכנס לנושא כללי (לפחות) לפרויקט, ולהתחיל להכין מצגת קצרה להצגה בשיעור.
אשלח בהמשך מסמך לתיאום זמני פגישות איתי בזמני השיעור.

- לאחר מכן נשוב לשיעורים פרונטליים, בהם כל צוות יציג הרצאה לא פורמלית קצרה (לא יותר מחצי שעה) שתדון ברקע הכללי לפרויקט (related work overview) ותציג התחלה של מסגרת כללית לפרויקט (ללא שום מימוש בשלב זה, רק כיוונים כלליים). במהלך תקופה זו יתכנו גם עוד הרצאות רגילות והרצאות אורח, מעבר רק להרצאות סטודנטים, לפי הזמן (עדכון בהמשך).

Main lecture topics

- Agents introduction - general and scientific
- Hypothesis generation
- End-to-end discovery agents